

AI based approach for scheduling Crop Rotation planning using Gale-Shapley algorithm

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Abstract

Crop rotation is the farming practice of growing different types of crops in the same area across a sequence of seasons, in order to improve soil health, optimize nutrients in the soil, reduce pest and disease pressure, and increase crop yields. Scheduling crop rotation planning is a complex problem, and farmers need to take into account many factors such as soil type, climate and crop requirements. In this paper we present an innovative AI based approach for scheduling Crop Rotation planning which is based on stable matching problems, and in particular on the Gale-Shapley algorithm, which is a well-known algorithm for solving this type of problems. The aim of the paper is to show that our approach is better than other naive algorithms in terms of obtained results, and to provide a detailed explanation of the approach we used.

Code — https://github.com/nardib/AI_Based_Crop_Rotation_Planning

Introduction

Crop rotation is a farming practice that involves growing different types of crops in the same area across a sequence of seasons. This practice is known since the ancient times and it is used to improve soil health, optimize nutrients in the soil, reduce pest and disease pressure, and increase crop yields. It's common knowledge that increasing the levels of chemical fertilizers and pesticides become necessary with monocultures to keep yields high while keeping bugs and disease at bay, so a well scheduled crop rotation can lead to better yields and a more sustainable farming system.

Over the years, many approaches have been proposed to schedule crop rotation planning, and they were mainly based on the experience and on some simple rules. Yet, crop rotation planning in general is a complex problem, and farmers need to take into account many factors such as soil type, climate and crop requirements. It is essential to find an effective approach to schedule crop rotation planning. This is especially important in the current scenario, where the world population is growing and requires to produce more food in a sustainable way.

Given the importance of the problem and its complexity, there are many attempts which aims at finding an efficient approach to schedule crop rotation planning using Artificial Intelligence (AI) techniques, which can help farmers to take better decisions and to optimize their crop rotation planning.

In literature, there are several articles that propose different approaches to schedule crop rotation planning and crop management using AI techniques. Some of them just propose ideas on how to implement software to help farmers in the decision-making process, like (Schöning and Richter 2021), while others discuss software implementations and the results obtained over simulated environments, like (Plant 1989) and (Pahmeyer, Kuhn, and Britz 2021). Some papers also provide algorithms and techniques to schedule the crop rotation, which use for example machine learning techniques, like (Fenz et al. 2023) who proposes a reinforcement learning approach to schedule the crop rotation, or local search metaheuristic techniques to optimize the crop rotation planning when the problem is expressed as a mathematical model, like (Chetty and Adewumi 2013), (Chetty and Adewumi 2014) and (Adewumi and Chetty 2017).

In this paper we present an innovative AI based approach for scheduling Crop Rotation planning, which is based on stable matching problems, and in particular on the Gale-Shapley algorithm.

The aim of the paper is to provide a detailed explanation of the approach we implement and to show that our approach is better than other naive algorithms in terms of obtained results.

Basic Notions

In this section, we are going to define some basic notions and to provide a formal definition of the Crop Rotation planning problem, in order to then apply the Gale-Shapley algorithm to find at first a stable matching between the crops and the fields, and then to schedule the crop rotation planning.

The problem of scheduling crop rotation planning can be formalized as a stable matching problem, where we have two equally sized sets, one for the crops and one for the fields. A stable matching is a matching between the elements of the two sets, where there are no two elements that would prefer to be matched with each other rather than with their current match. In our case, we can define the preferences of the crops and the fields based on their characteristics, such as soil type, climate and crop requirements.

The rotation planning can be achieved by finding a sequence of stable matchings, which take into account the changes in the preferences of the crops and the fields over time, due to factors such as weather condition and changes

in the soil health.

In order to define dynamic preferences, which are the preferences that change over time, we can use functions that takes into account the current state of the fields and the crop characteristics, and that updates the preferences of the crops and the fields accordingly. This allows us to adapt the rotation planning to unexpected events, such as changes in the water availability of a specific field.

In our discussion, we will take into account only a limited number of factors to define the preferences between the two sets, which are:

- Water demand and availability: some crops require more water than others, and some fields have more water available than others.
- Nitrogen demand and availability: some crops require more nitrogen than others, and some fields have more nitrogen available than others.
- Sun exposure: some crops require more sun exposure than others, and some fields have more sun exposure than others.
- pH level: some crops require a specific pH level, and some fields have a specific pH level. This is important because some crops can only grow in specific pH levels, and some fields have a specific pH level that can be suitable for certain crops.

A higher score between a crop and a field imply a stronger compatibility.

The function used to define the preferences of the crops follows the following rules:

- If the pH level of the field is not suitable for the crop, then the crop will have the lowest possible score for that field, regardless of the other factors.
- For sun, water and nitrogen, a score is assigned by computing the distance between the demand of the crop and the availability of the field; the score is higher when the distance is smaller, and it is lower when the distance is bigger.

The function used to define the preferences of the fields follows the following rules:

- If the family of the crop is the same as the one of the crop that was planted in the field in the previous season, then the field will have the lowest possible score for that crop, regardless of the other factors. This is because planting crops from the same family in consecutive seasons turns out to be terrible for the soil health, and it can lead to a decrease in yields.
- If the crop to be planted is a legume, then the score is higher, because legumes augments nitrogen in the soil, which can be beneficial for the next crop that will be planted in the field.
- If the nitrogen demand of the crop is higher than the field availability, then the score is lower, because it can lead to a decrease in yields and in soil health.
- Water demand is also taken into account in the same way as it is taken into account for the crops, because it can lead to a decrease in yields and in soil health.

It's easy to see that this framework can be applied also to more complex scenarios: we take into account more factors to define the preferences of the crops and the fields, and we consider larger sets of crops and fields.

The main idea is to use the Gale-Shapley algorithm to find a stable matching between the crops and the fields, and to update the preferences of the crops and the fields over time, in order to adapt the rotation planning to unexpected events. In the end, we simulate the crop rotation planning over numerous seasons, to establish the total yield produced by all the fields, in order to prove the effectiveness of our approach.

Dataset and implementation

In order to evaluate the efficiency of our approach, we created a dataset of 10 crops and 10 fields. The dataset was created after a brief research on the characteristics of the crops and the fields, so it is meant to be used just for testing the algorithm and to show how it works, and it is not meant to be used for real-world applications¹.

The values for water demand/availability, nitrogen demand/availability and sun demand/exposure are all normalized in order to have a value in $[0, 1]$.

Tab. 1 and Tab. 2 show a summary of the characteristics used for testing the algorithm².

Name	Water	Nitrogen	Sun	pH Min-pH Max
Wheat	0.5	0.6	0.8	6.0-7.5
Corn	0.8	0.9	1.0	5.8-7.0
Soybean	0.6	0.3	0.9	6.0-7.0
Potato	0.7	0.7	0.8	5.0-6.5
Tomato	0.7	0.8	1.0	5.5-7.0
Alfalfa	0.6	0.2	0.9	6.5-7.5
Barley	0.4	0.5	0.8	6.0-7.5
Sunflower	0.5	0.6	1.0	6.0-7.5
Pea	0.5	0.3	0.7	6.0-7.5
Canola	0.6	0.7	0.9	5.5-7.0

Table 1: Crop characteristics

The preferences from the crop set to the field set are defined by the following function:

`crop_to_field_score(crop, field)`

$$c(f) = \begin{cases} 0 & \text{if pH} = 0 \\ 0.30 \cdot W + 0.25 \cdot S + 0.25 \cdot N + 0.20 \cdot \text{pH} & \text{otherwise} \end{cases} \quad (1)$$

The function evaluates the similarity between crop demands and field availability on different factors: pH represents the normalized distance between crop's pH level required and field's pH level, W is the normalized distance between crop's water demand and field's water availability,

¹This does not affect the validity of the test results, since the preferences are dynamically adapted based on the characteristics given to the program.

²The real dataset we used also contains the family of the crops and the crop that was planted in each field in the previous season, which are used to define the preferences of the fields, but they are not shown in the tables for the sake of simplicity

Name	Soil Type	pH	Water	Sun	Nutrients
Field1	loam	6.8	0.7	0.9	0.8
Field2	sandy	6.2	0.4	1.0	0.5
Field3	clay	7.2	0.8	0.8	0.9
Field4	loam	6.5	0.6	0.7	0.7
Field5	silty	6.9	0.7	0.9	0.8
Field6	sandy	5.8	0.3	1.0	0.4
Field7	clay	7.0	0.9	0.8	0.9
Field8	loam	6.3	0.5	0.8	0.6
Field9	silty	6.7	0.6	0.9	0.7
Field10	loam	7.1	0.7	1.0	0.8

Table 2: Field characteristics

and the same reasoning is applied for the calculation of S , for the sun, and N , for the nutrients. If the pH level of the field is not suitable for the crop, then the crop has a score for that field of 0, regardless of the other factors.

The preferences from the field set to the crop set are defined by the following function:

```
field_to_crop_score(crop, field)
```

$$f(c) = \begin{cases} 0 & \text{if cr_fam} = \text{lastCr_fam} \\ W + N + 0.4 \cdot \text{is_legume} & \text{otherwise} \end{cases} \quad (2)$$

As in the crop’s preferences function, it evaluates the similarity between field availability and crop demands on different factors, like water (W) and nutrients (N), but we also consider the *is_legume* flag, that contributes to an higher score for that crop. Furthermore, if the current crop family is the same as the one of the last planted crop, then the score of the crop is set to 0, regardless of other factors.

After having defined the score rules, we used the `generate_preferences(crops, fields)` function to create a preference list for each crop and field, in descending order of score, so that the first item is the most preferred one. For example:

```
Corn -> ['Field10', 'Field3', 'Field1', ...]
Field4 -> ['Soybean', 'Alfalfa', 'Pea', ...]
```

The Figure. 1 shows the preference values computed for each crop to each field, while the Figure. 2 shows the preference values computed for each field to each crop, for the data in the datasets above. The values are normalized in order to have a value in $[0, 1]$.

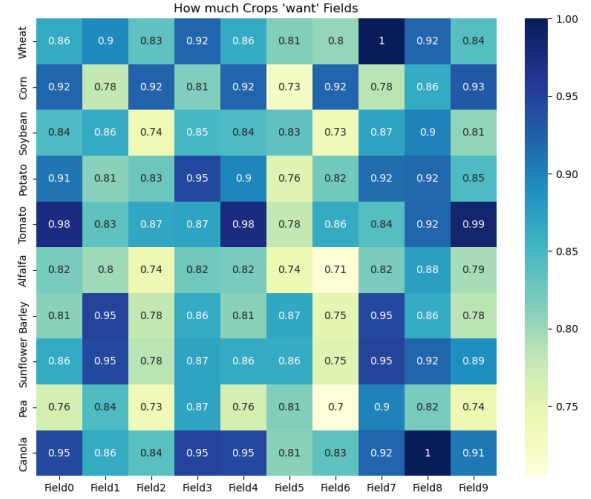


Figure 1: Example of preferences matrix from crops to fields



Figure 2: Example of preferences matrix from fields to crops

Finally, the stable matching between crops and fields is performed by applying the Gale-Shapley algorithm, implemented in the function `gale_shapley(crop_prefs, field_prefs)`.

In our implementation, the Gale-Shapley algorithm outputs a crop assignment for each field, representing a single-season allocation. Since the overall planning spans multiple seasons, the algorithm is applied iteratively for each season to construct the complete multi-year plan.

Evaluation

To assess the performance of our approach, we simulate the crop rotation using the `update_field(field, crop)` function, which updates the state of each field based on the crop planted in the previous season. The update rules are defined as follows:

$$\text{last_crop_family} \leftarrow \text{crop.family} \quad (3)$$

$$\text{nutrient_level} \leftarrow \begin{cases} +0.2 & \text{if is_legume} \\ -0.15 \cdot \text{nutrient_demand} & \text{otherwise} \end{cases} \quad (4)$$

$$\text{water_availability} \leftarrow +\mathcal{U}(-0.1, +0.1) \quad (5)$$

The value of the water availability for each field is updated using a value sampled from a uniform distribution between $[-0.1, 0.1]$ in order to simulate the uncertainty given by the weather on the fields. All the updated values are then clamped to have a value between $[0, 1]$.

To provide a measure of the performances of the algorithm we evaluate the yield for each field using the function `compute_yield(crop, field)` and sum all the values to compute the total seasonal yield. Higher total yields indicate a better crop rotation.

To calculate the field’s yield, the following function is used:

$$y(c) = \text{product} \cdot \text{compatibility} \cdot N_f \cdot W_f \cdot S_f \quad (6)$$

where

- *product* is the crop productivity; it is set to 1 for all crops, except for corn which is set to 1.2 and soybean to 0.9.
- *compatibility* is the compatibility between the field and the crop; it represents the crop to field score.
- *N_f* is the nutrient factor and it’s calculated as the nutrient level of the field multiplied by 1.5.
- *W_f* is the water factor, representing the water stress of the field given by the crop. It’s calculated as the normalized difference between water demand and water availability, multiplied by 1.5.
- *S_f* is the sun factor and it’s normalized difference between sun demand and sun availability.

To benchmark our approach, we compare it against three baseline methods: a *random* algorithm, which assigns crops to fields uniformly at random; a *greedy* algorithm, which assigns each crop to the field with the highest crop-to-field compatibility score; and a modified version of the Q-learning algorithm proposed by (Fenz et al. 2023). The first two baselines represent naive approaches that do not account for the dynamic evolution of field states or adapt the plan in response to unexpected events.

In the reinforcement learning literature, Q-learning is a model-free algorithm in which an agent learns an optimal policy by iteratively estimating the Q-values for each state-action pair (s, a) (Sutton and Barto 2018). The standard update rule is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \cdot \left(r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a) \right)$$

where r is the immediate reward, s' is the next state, α is the learning rate, and γ is the discount factor.

To adapt this framework to our setting, we encode the state s as a compact 3-tuple comprising the botanical family of the last crop planted in the field, the current nutrient level, and the current water availability. The action a corresponds

to the selection of the crop to plant, while the reward r is derived from the resulting crop yield and its contribution to the overall seasonal performance.

In our implementation, each field is treated as an independent agent with its own Q-table, indexed by (s, a) pairs. Since not all crops are admissible for every field at every season, the algorithm first filters out crops belonging to the same botanical family as the previously planted crop. Crop selection follows an ϵ -greedy policy: with probability ϵ a random valid crop is chosen, and with probability $1 - \epsilon$ the crop with the highest Q-value is selected. To promote exploration in the early stages of training, ϵ is initialized to 1 and annealed after each episode until it reaches a minimum of 0.02.

Training is organized into episodes, each consisting of multiple seasons. At every season, the current field state is encoded, a crop assignment is generated, yields are computed, and the field states are updated for the following season. Since multiple fields may prefer the same crop, conflicts are resolved by ranking proposals according to learned Q-values combined with a fairness rotation term designed to prevent systematic bias toward the same fields.

The reward signal used to update the Q-table combines multiple components: a season-level normalized reward, a differential term comparing the chosen action against a baseline alternative, and an advantage term measuring the relative performance of the field with respect to the average yield of all other fields. This reward shaping improves the stability of the learning process and better aligns the local optimization of each field with the global objective of maximizing total rotation yield.

To mitigate the variance introduced by random exploration, we train an ensemble of independent Q-learners, each initialized with a different random seed. During evaluation, the Q-values of all learners are averaged, and the resulting scores are used to determine the final crop assignment for each field. When available, the Hungarian algorithm³ is applied to find the one-to-one field-crop assignment that maximizes the total averaged Q-value. This ensemble strategy reduces sensitivity to any single training run and generally improves robustness.

Results

The experiment consists of simulating crop rotation over a period of 10 seasons and measuring the total yield produced across all fields.

Method	Total yield	Mean yield	
		per season	Std per season
Q-learning	39.721730	3.972173	0.334039
Gale-Shapley	38.787668	3.878767	0.386000
Greedy	32.794923	3.279492	0.646843
Random	30.787130	3.078713	0.394925

Table 3: Summary of results across all methods

³The Hungarian method is a combinatorial optimization algorithm that solves the assignment problem in polynomial time.

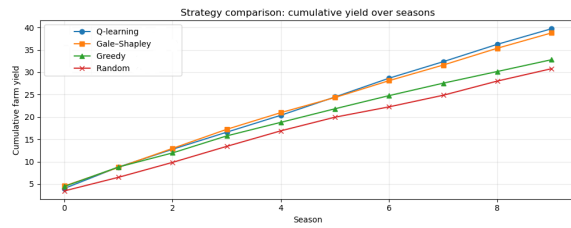


Figure 3: Cumulative total yield over 10 seasons for each algorithm

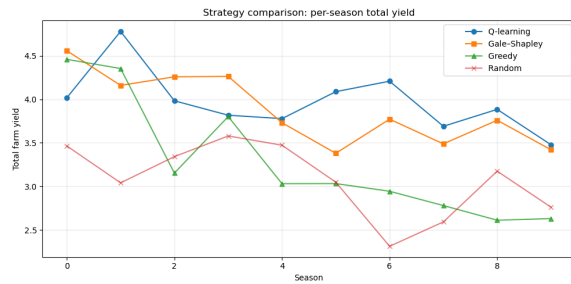


Figure 4: Seasonal yield over 10 seasons for each algorithm

Season	Season 0	Season 1	Season 2	Season 3	Season 4	Season 5
Field						
Field0	Potato	Alfalfa	Tomato	Canola	Alfalfa	Tomato
Field1	Barley	Sunflower	Sunflower	Pea	Barley	Pea
Field2	Corn	Corn	Wheat	Wheat	Sunflower	Soybean
Field3	Alfalfa	Potato	Pea	Barley	Corn	Corn
Field4	Canola	Soybean	Potato	Alfalfa	Canola	Sunflower
Field5	Sunflower	Pea	Barley	Sunflower	Pea	Barley
Field6	Wheat	Tomato	Corn	Potato	Soybean	Potato
Field7	Pea	Wheat	Canola	Soybean	Potato	Wheat
Field8	Soybean	Barley	Alfalfa	Corn	Tomato	Canola
Field9	Tomato	Canola	Soybean	Tomato	Wheat	Alfalfa

Figure 5: Gale-Shapley crop rotation plan over 6 seasons

Season	S0	S1	S2	S3	S4	S5
Field						
Field0	Soybean	Corn	Soybean	Tomato	Alfalfa	Tomato
Field1	Barley	Pea	Tomato	Sunflower	Soybean	Barley
Field2	Canola	Tomato	Alfalfa	Corn	NaN	Canola
Field3	Pea	Potato	Pea	Potato	Pea	Potato
Field4	Wheat	Alfalfa	Canola	Alfalfa	Canola	Alfalfa
Field5	Sunflower	Barley	Sunflower	Wheat	Sunflower	Wheat
Field6	Corn	Soybean	Corn	Canola	Potato	Corn
Field7	NaN	Wheat	Potato	Pea	Barley	Pea
Field8	Alfalfa	Canola	Wheat	Soybean	Wheat	Soybean
Field9	Tomato	Sunflower	Barley	NaN	Tomato	Sunflower

Figure 6: Q-learning crop rotation plan over 6 seasons

As shown in Figure 3 and Table 3, each method exhibits distinct characteristics. In this simulation, Q-learning achieves the highest total yield; however, Gale-Shapley re-

mains the preferable choice when interpretability and controllability are key requirements. Gale-Shapley relies on explicit preference functions defined over both crops and fields, where each term has a clear meaning — water availability, nutrient levels, sun exposure, crop family, and legume presence. As a result, every decision is fully transparent: a farmer can inspect, justify, and modify any component of the preference function to incorporate domain knowledge or new agronomic constraints.

Q-learning, by contrast, attempts to maximize total yield by inferring soil preferences and crop effects through trial and error. Obtaining reliable Q-value estimates requires extensive exploration of crop-field combinations across a large number of seasons, which can be both time-consuming and computationally expensive. Furthermore, the learned Q-values are not easily interpretable: while they encode expected future rewards, they do not provide a clear rationale for why a specific crop was selected for a given field. With limited training data, Q-learning may also exhibit high variance and poor generalization.

In general, the results obtained are variable and different, because we added a randomness factor to simulate the uncertainty given by the weather on the fields, and this leads to different results across different runs.

By repeating the training process across multiple independent runs, we observe that Gale-Shapley delivers more consistent performance, making it a safer and more practical choice in settings which requires explainability and a difficult and costly collection of data.

Conclusions

The approach for scheduling Crop Rotation planning presented in this paper is based on the Gale-Shapley algorithm, so it exploits the properties of the crops and the fields to determine the matching. This characteristic allows the approach to achieve both an higher yield and an improvement of the fields' status and soil health. In real world applications, it can be used as a decision-making support system for farmers to reach a more sustainable farming system.

We have shown that our approach achieves good results in terms of singular and total yield, but a precious advantage is the explainability: it gives the possibility to the farmers to look at the values of the preferences and also to understand the reasons behind the crop assignment. As a consequence, it can be easily modified to take into account more factors to define the preferences of the crops and the fields, or to consider a larger sets of crops and fields.

Furthermore, another advantage with respect to other approaches is that it does not require a training phase and a large amount of data to learn the preferences of the crops and the fields. This makes our approach more robust to changes in the environment and to unexpected events, such as changes in the water availability of a specific field.

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